IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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https://github.com/kenmarciel/space-race

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology

- Data collection methodology:
  - Get request to Space X API, and web scraping of Wikipedia page for Falcon 9 launches
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Executive Summary

Results

- Launch Success Rate
  - Generally increases with number of launches at a site
  - Highest for payloads between 1,500 and 4,000 kg
  - Highest for ES-L1, GEO, HEO, SSO, and VLEO orbits
  - Highest for Kennedy Space Center Launch Complex 39A
- Machine learning
  - Decision tree model performed best with 94% classification accuracy

Introduction

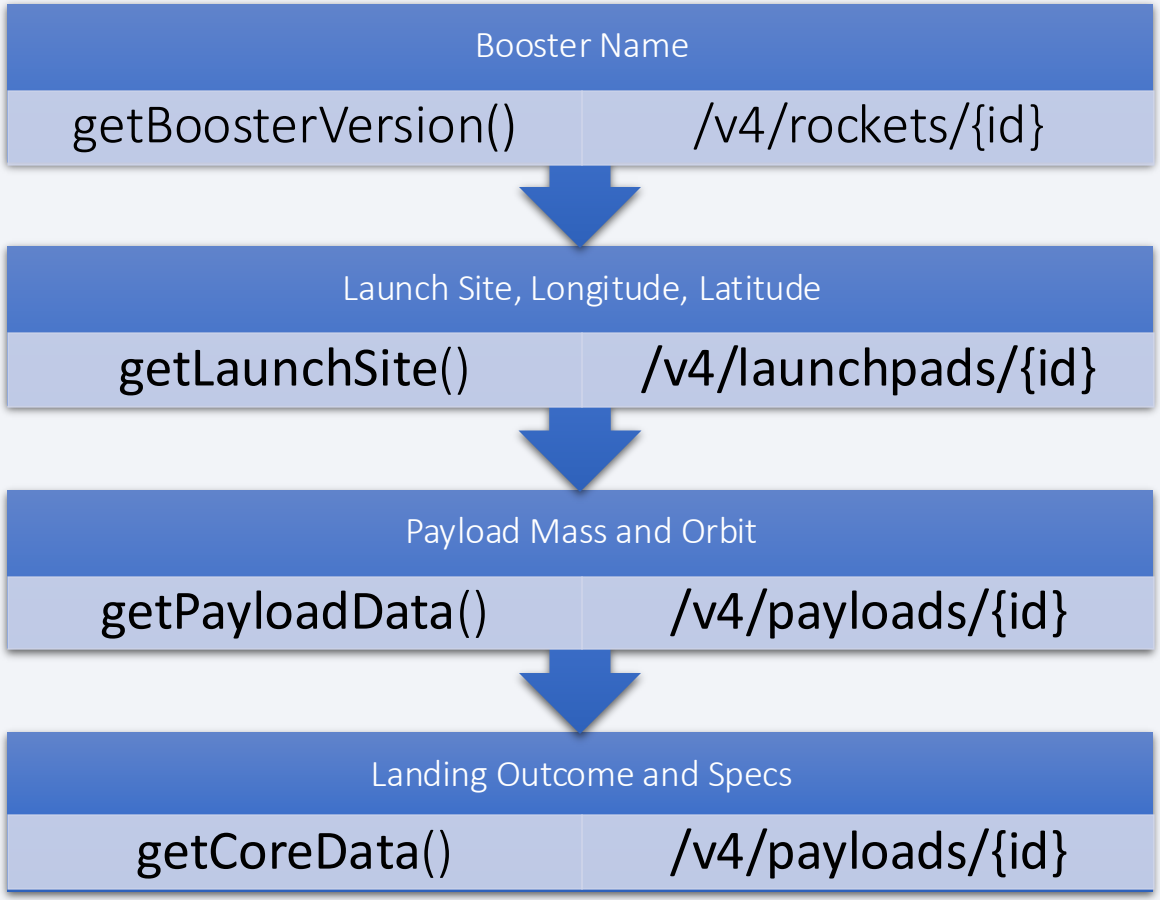
- Space X website advertises Falcon 9 rocket launches
  - Cost of $62 million each
  - Other providers cost upward of $165M each
  - Much of savings is because Space X can reuse the first stage
- Successful landing of first stage can determine cost of launch
  - Useful information for company that wants to bid against Space X
- Purpose of this analysis is to predict if first stage will land
  - Based on data from previous Falcon 9 launches

Section 1

Methodology

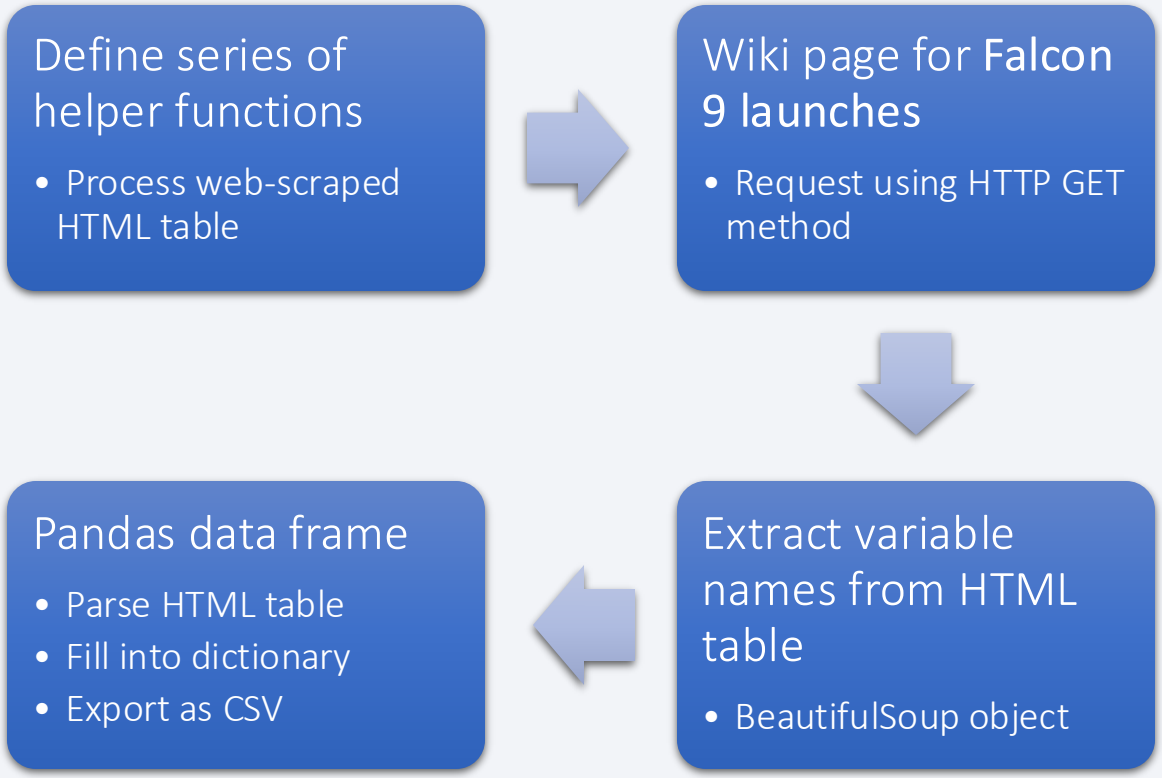
Data Collection – SpaceX API

- Create series of helper functions
- Request to the SpaceX API
  - response = requests.get(spacex_url)
- ID numbers in launch data
  - Extract information
- Clean the requested data
- https://github.com/kenmarciel/space-race/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



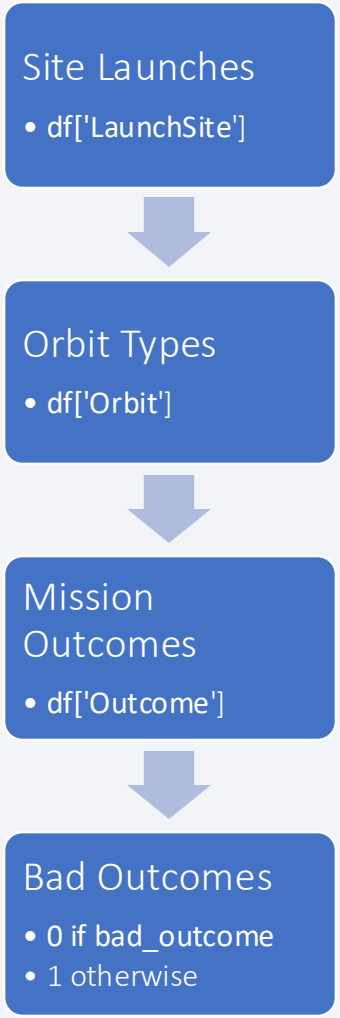
Data Collection - Scraping

- Extract Falcon 9 launch records
  - HTML table from Wikipedia
- Load into data frame
  - Export as CSV file
- https://github.com/kenmarciel/space-race/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

- Calculate counts for launch data
  - Apply value_counts() method to columns
- https://github.com/kenmarciel/space-race/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

- Pairwise effect of variables on Launch Outcome
  - Flight Number vs. Payload Mass
  - Flight Number vs. Launch Site
  - Flight Number vs. Orbit Type
  - Payload Mass vs. Launch Site
  - Payload Mass and Orbit Type
- Relationship between Success Rate and Orbit Type
  - Identify orbits with highest success rates
- Launch Success Yearly Trend
  - Change in average annual success rate over time
- https://github.com/kenmarciel/space-race/blob/main/edadataviz.ipynb

EDA with SQL

- Display
  - Names of unique launch sites in space mission
  - 5 records where name of launch site begins with 'CCA'
  - Total payload mass carried by boosters launched by NASA (CRS)
  - Average payload mass carried by booster version F9 v1.1
- Rank
  - Landing outcome by count between 6/4/2010 and 3/20/2017
- List
  - Date when first successful landing outcome in ground pad was achieved
  - Boosters with success in drone ship and payload mass between 4,000 and 6,000 kg
  - Total number of successful and failure mission outcomes
  - Booster versions carrying maximum payload mass
  - Month, landing outcome, booster version, and launch site for drone ship in 2015
- https://github.com/kenmarciel/space-race/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Circle object and popup label for each launch site based on its coordinates
  - Zoom in/out to assess proximity to coast or equator
- Cluster object for launch outcomes at each launch site
  - Identify which launch sites have relatively high success rates
- Mouse position for coordinates of moving cursor on map
  - Find coordinates of points of interest
- Distance marker and line from launch site to points of interest
  - Proximity to nearest railway, highway, coastline, city, etc.
- https://github.com/kenmarciel/space-race/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Launch Site Drop-down Input Component
  - See which launch site has largest success count
  - Select a site and check its detailed success rate (class=0 vs. class=1)
- Pie chart of launch site success rate
  - Visualizing launch success counts based on site selected
- Range slider to select payload
  - See if payload is correlated to mission outcome
- Scatter plot of payload versus launch outcome and booster version
  - Correlation of payload with mission outcome and booster version
- https://github.com/kenmarciel/space-race/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

Create binary outcome variable
- Success or failure of each launch

Standardize variables
- Mean of zero and unit variance

Randomly split data
- 80% as training set
- 20% as test set

Find best model hyperparameters
- Logistic regression, SVM, decision tree, KNN

Find method that performs best
- Based on cross-validation score and confusion matrix

- https://github.com/kenmarciel/space-race/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

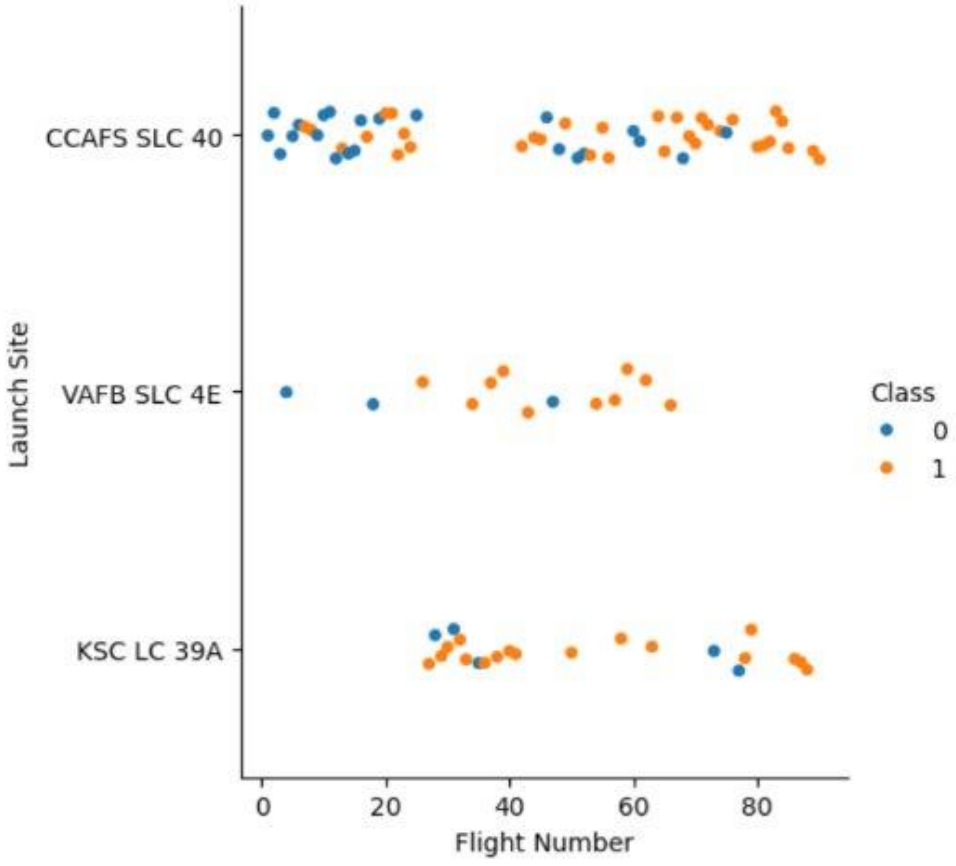
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

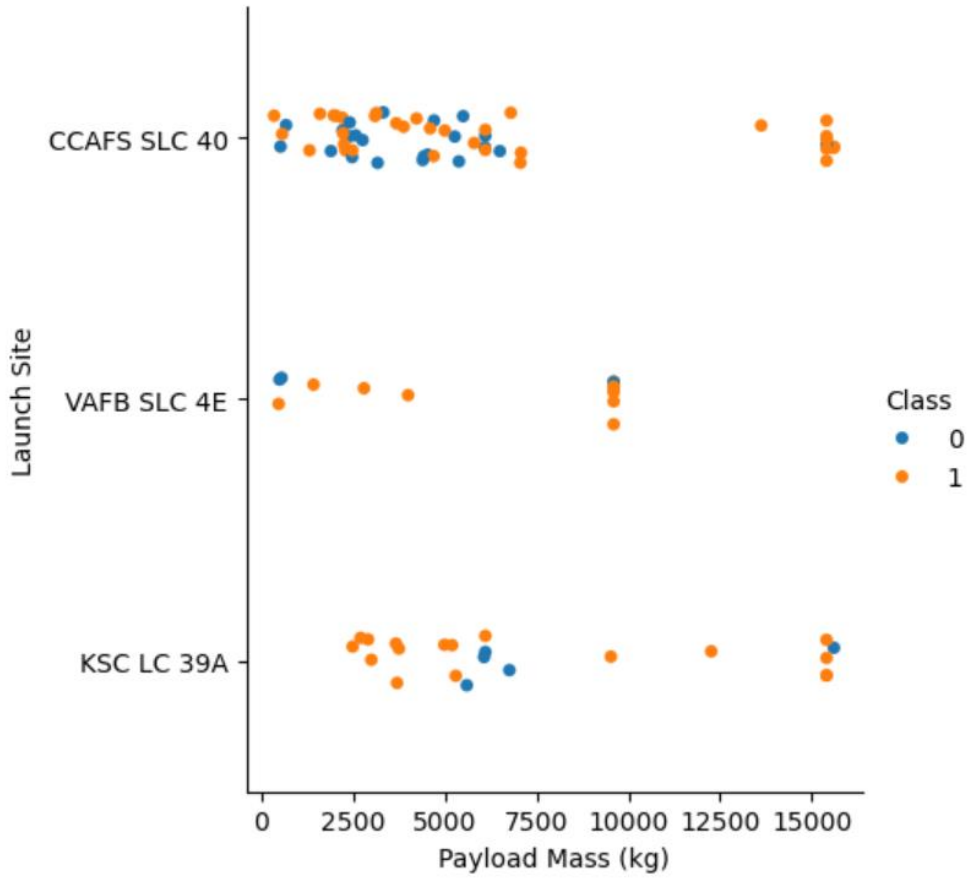
Insights drawn from EDA

Flight Number vs. Launch Site



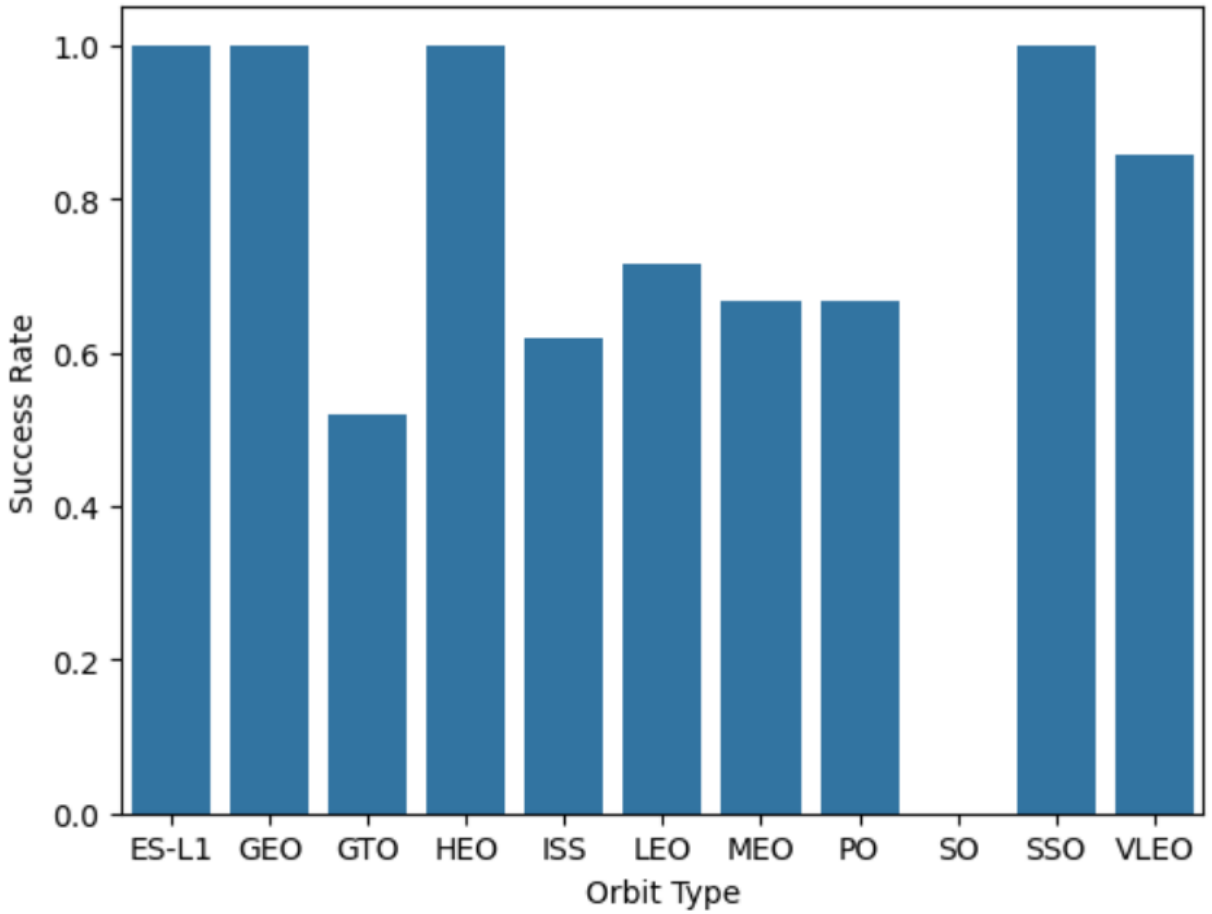
Success rate generally improves with number of flights at launch sites CCAFS SLC 40 and VAFB SLC 4E, which cannot be said of launch site KSC LC 39A.

Payload vs. Launch Site



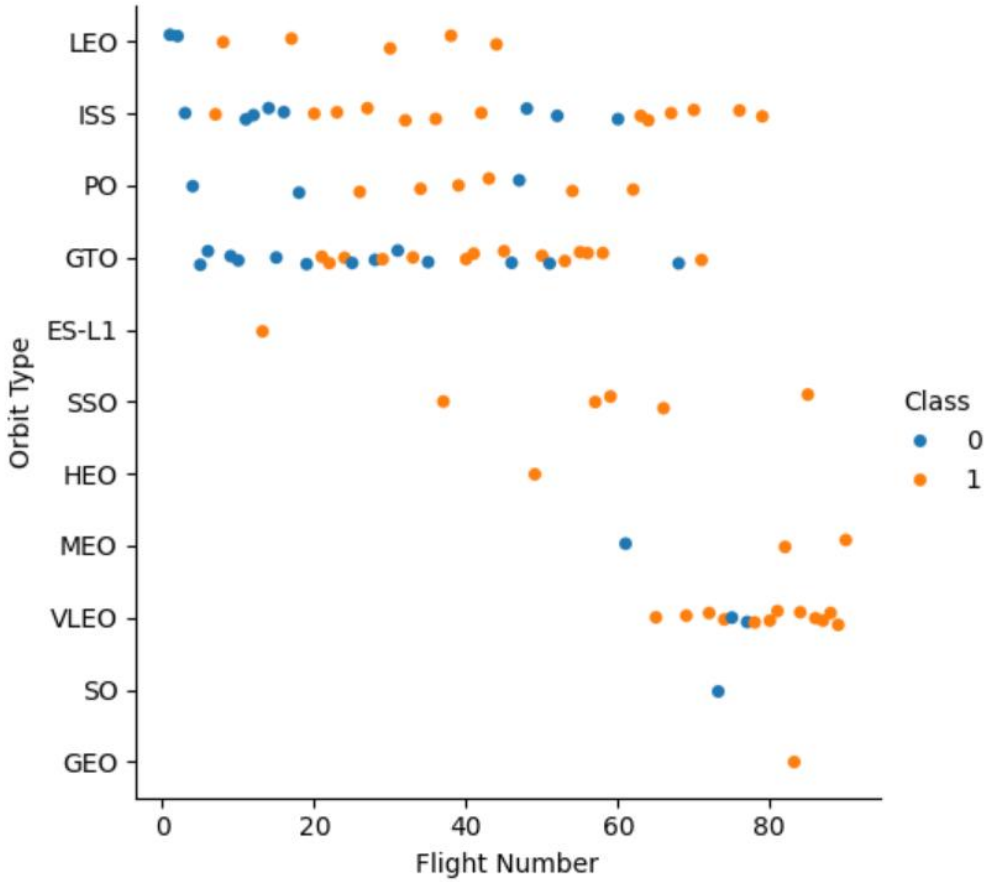
Success rate generally improves as payload increases. For the VAFB-SLC launchsite there are no rockets launched for heavy payload mass(greater than 10000).

Success Rate vs. Orbit Type



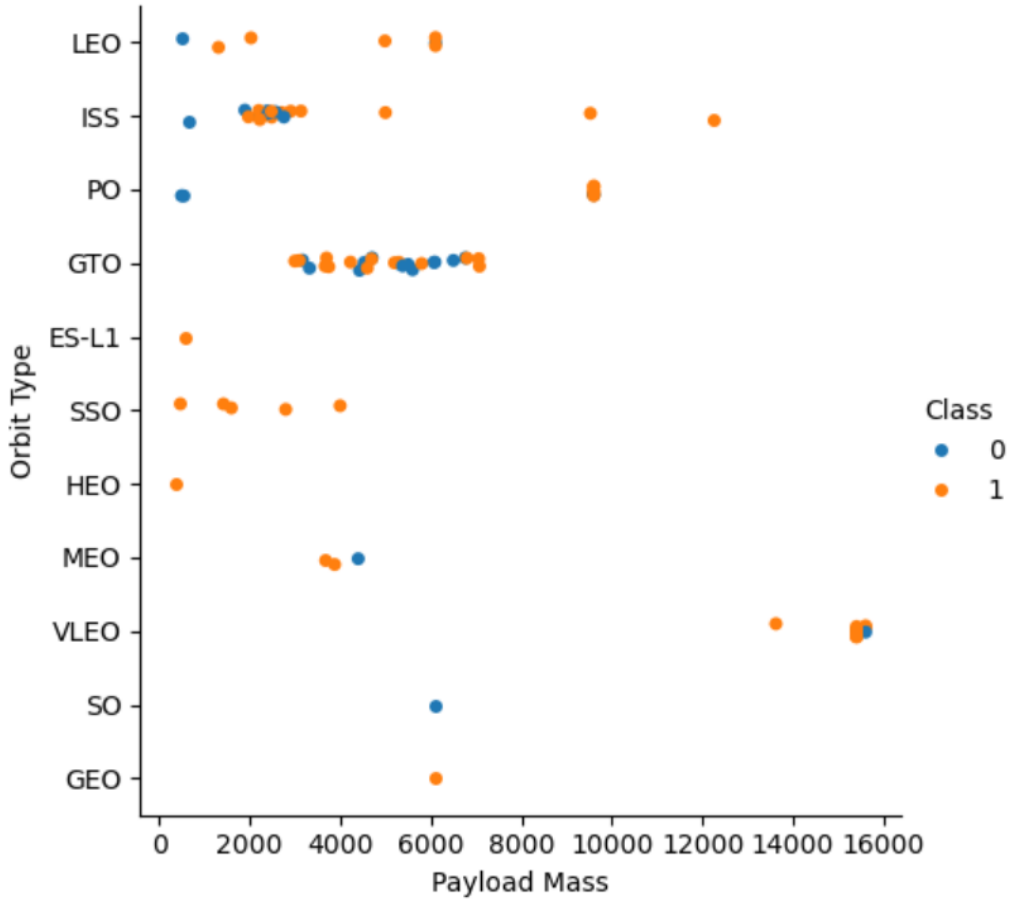
The orbits with the highest success rates are ES-L1, GEO, HEO, SSO, and VLEO.

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

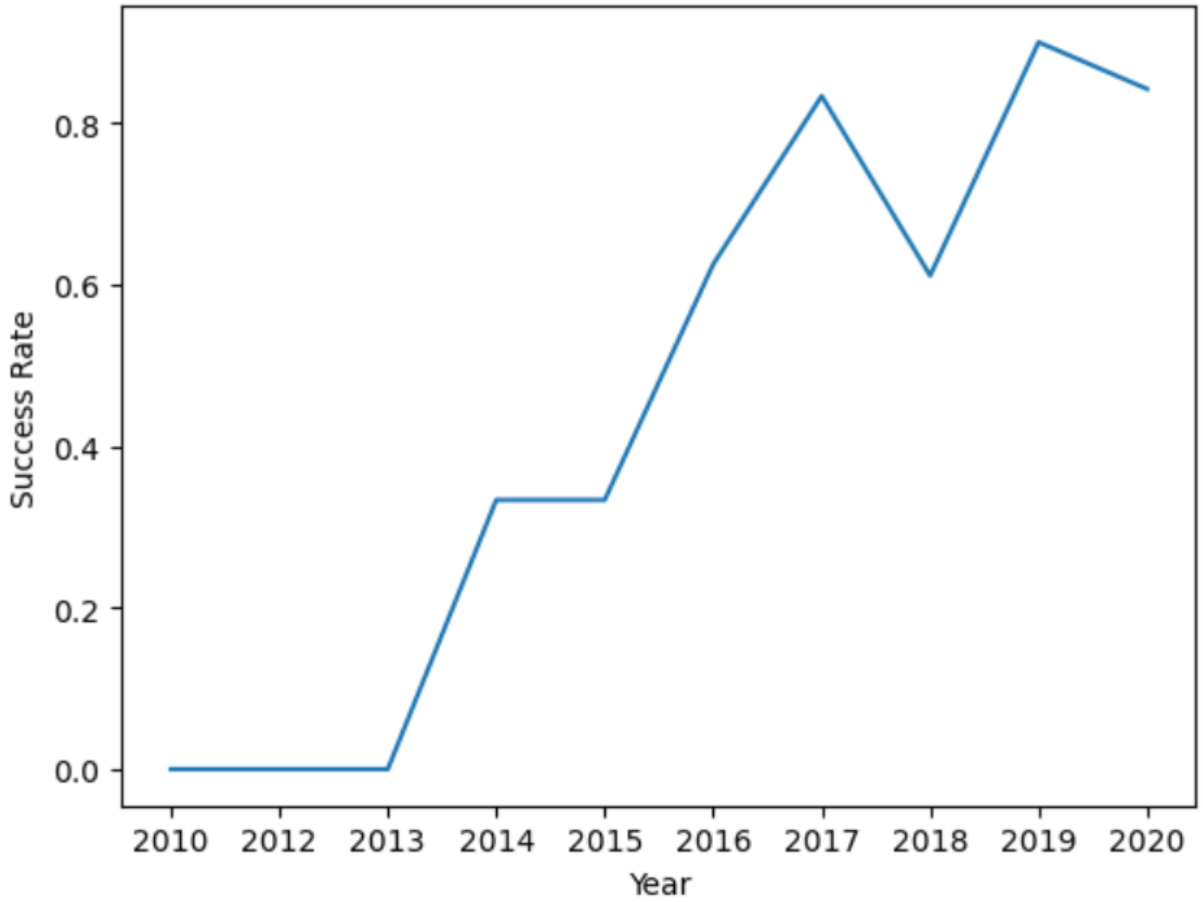
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



You can observe that the success rate since 2013 kept increasing until 2020.

All Launch Site Names

- Names of the unique launch sites in the space mission:
  - CCAFS LC-40
  - VAFB SLC-4E
  - KSC LC-39A
  - CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- First five records in dataset where name of launch site begin with 'CCA':

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload mass carried by boosters launched by NASA (CRS):
    - 45,596 kg

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1:
    - 2,928.4 kg

First Successful Ground Landing Date

- Date of first successful landing outcome on ground pad:
  - December 22, 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4,000 but less than 6,000 kg:

  - F9 FT B1022
  - F9 FT B1026
  - F9 FT B1021.2
  - F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes:

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Booster versions which have carried the maximum payload mass:

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

2015 Launch Records

- Failed landing outcomes in drone ship, their booster versions, and launch site names for year 2015:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between 6/4/2010 and 3/20/2017, in descending order:

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

SpaceX Launch Sites

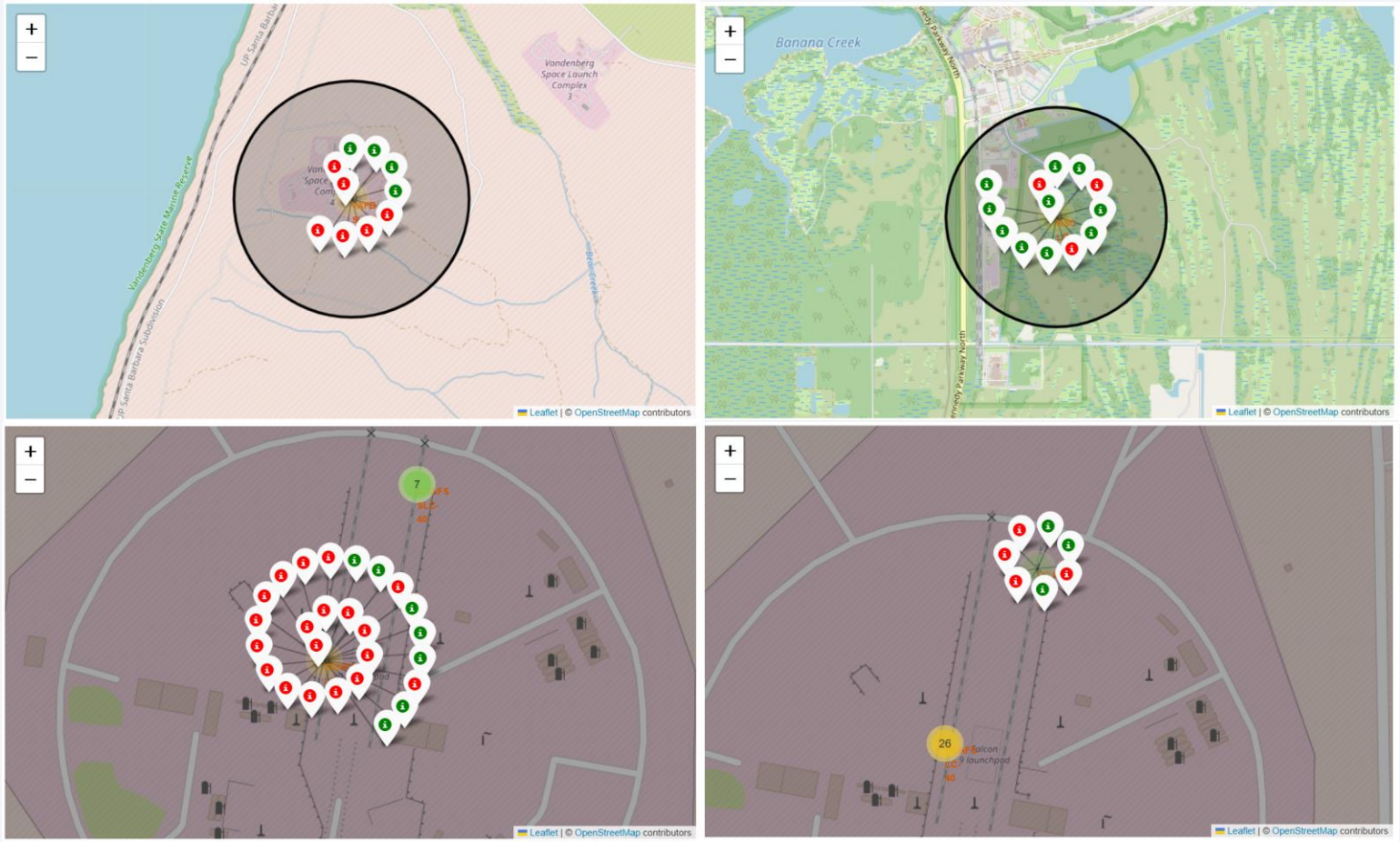
CCAFS LC-40 and CCAFS SLC-40 are located in Cape Canaveral, Florida

KSC LC-39A is located in Merritt Island, Florida

VAFB SLC-4E is located in Lompoc, California

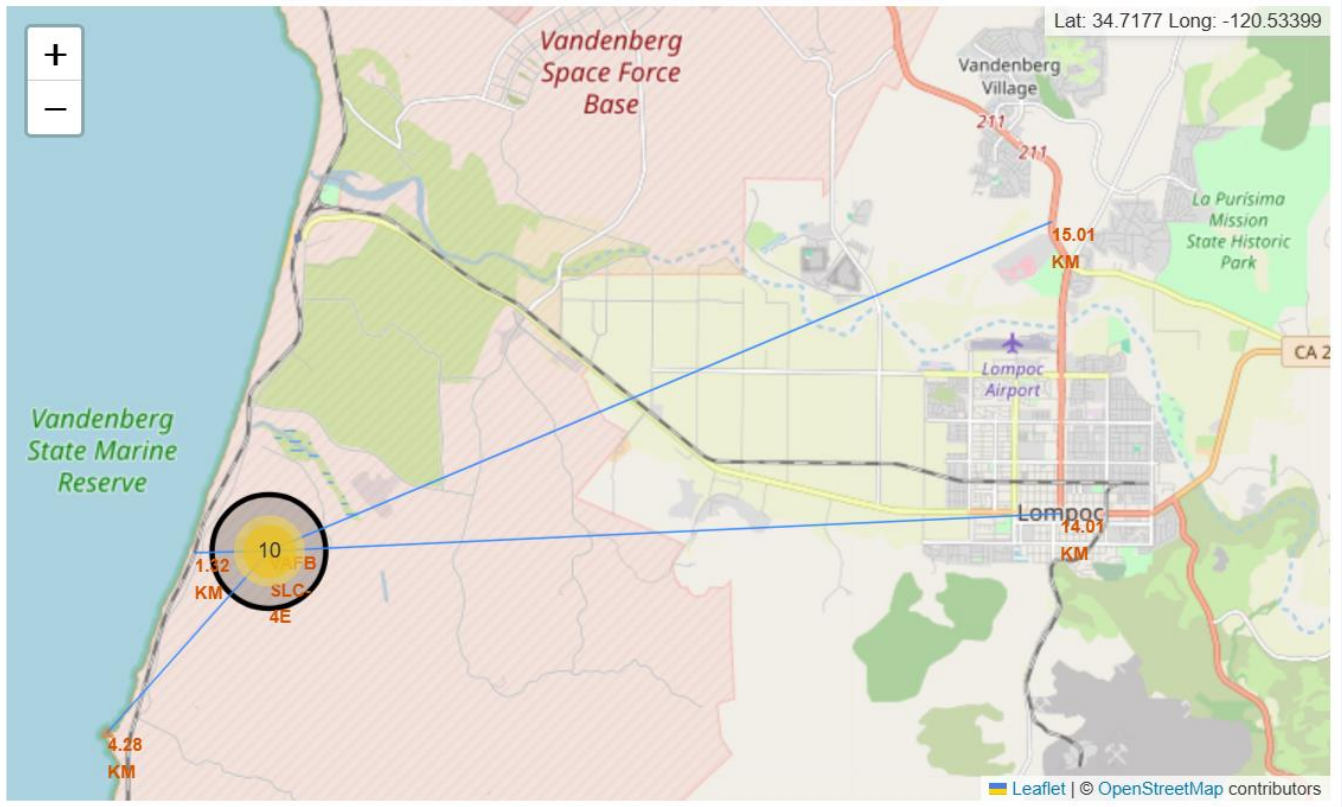
Outcomes at Each Launch Site

- Launch Outcomes
  - Green = Success
  - Red = Failed
- Launch Sites
  - Upper Left
    - VAFB SLC-4E
  - Upper Right
    - KSC LC-39A
  - Lower Left
    - CCAFS LC-40
  - Lower Right
    - CCAFS SLC- 40



Distance between VAFB SLC-4E and Proximities

- Point of interest
  - Promontory 35 m
  - 4.28 km
- Closest railway
  - UP Santa Barbara Subdivision
  - 1.32 km
- Closest highway
  - Cabrillo Highway
  - 15.01 km
- Closest city
  - Lompoc
  - 14.01 km

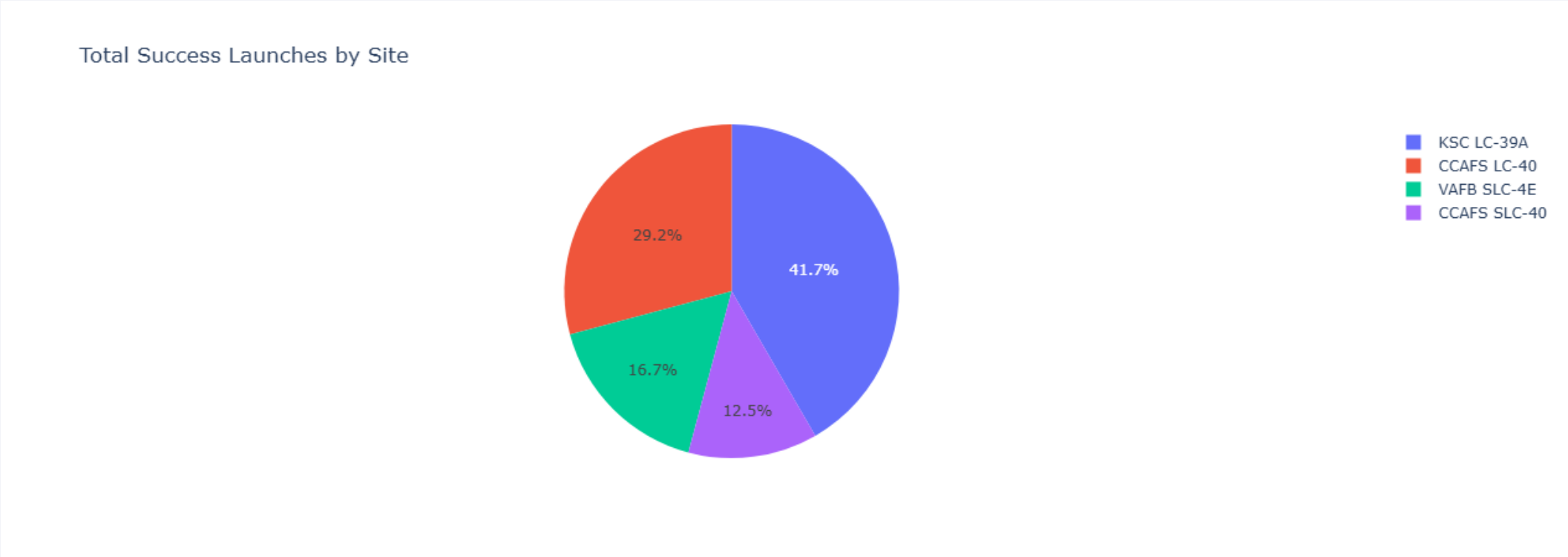


Section 4

Build a Dashboard with Plotly Dash

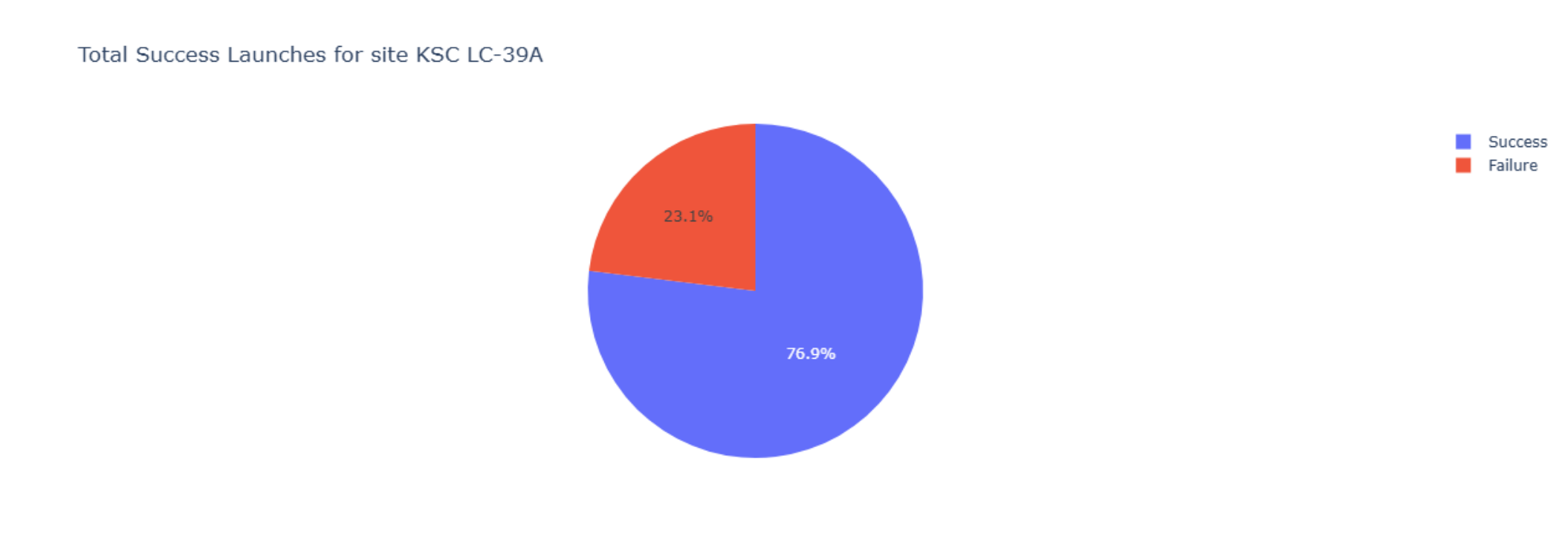
Launch Success Count for All Sites

- Proportion of total successful launches from each launch site



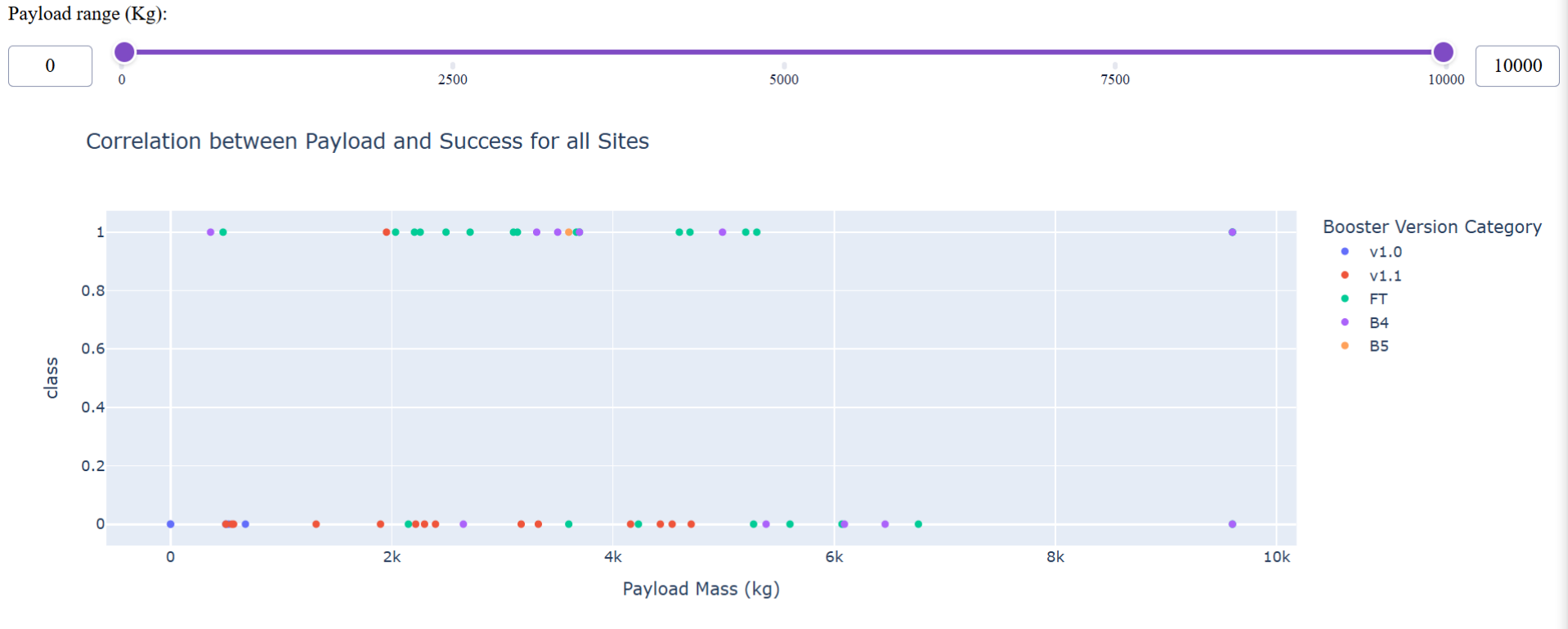
Launch Success Count for Site KSC LC-39A

- Ratio of successful to failed launches for site KSC LC-39A



Payload vs. Launch Outcome for All Sites

- Success rate is highest for payloads between 1,500 and 4,000 kg
- FT is the booster version with the highest success rate overall

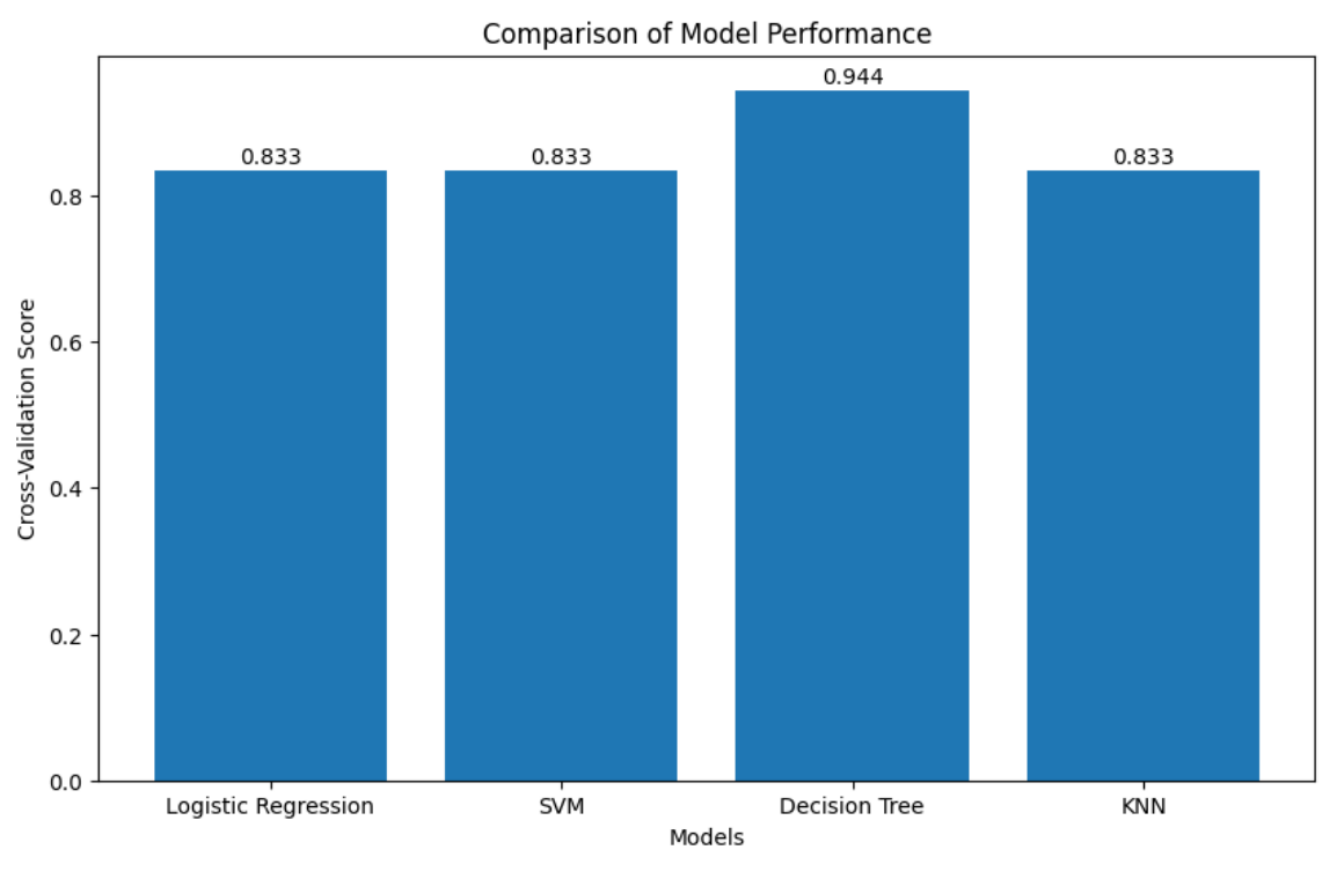


Section 5

Predictive Analysis (Classification)

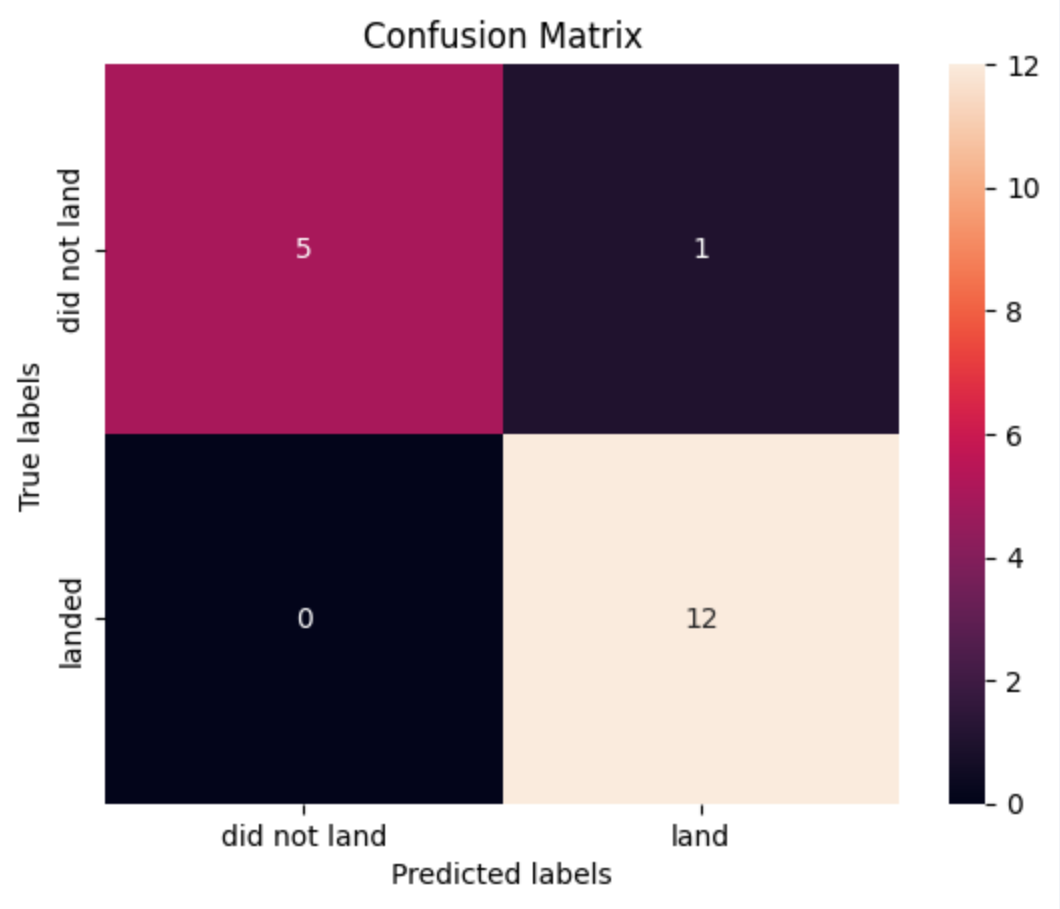
Classification Accuracy

The model that performed best is the decision tree, with an accuracy score of 94.4%.



Confusion Matrix

The decision tree correctly predicted all 12 successful launches, and correctly predicted five of the six failed launches.



Conclusions

- Launch success rate
  - Generally increases with number of launches at a site
  - Generally increases in direct proportion to payload mass
  - Highest for payloads between 1,500 and 4,000 kg
  - Highest for ES-L1, GEO, HEO, SSO, and VLEO orbits
  - Highest for Kennedy Space Center Launch Complex 39A
- Machine learning
  - Decision tree model performed best with 94% classification accuracy
- The above findings suggest the optimal approach for an alternate company to bid against Space X for a rocket launch

Appendix

- Space X API
  - https://api.spacexdata.com/v4/launches/past
- Wikipedia page for list of Falcon 9 and Falcon Heavy launches
  - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Thank you!